

7. (1.75 points)

Given the following information:

<u>Cumulative Reported Losses (\$000)</u>		
<u>Accident</u> <u>Year</u>	<u>As of 24</u> <u>Months</u>	<u>As of 36</u> <u>Months</u>
2009	3,000	5,000
2010	5,000	8,000
2011	2,500	9,000
2012	3,200	6,000
2013	3,800	7,000
2014	2,500	

a. (1.25 points)

Create a scatter plot of the weighted residuals for reported losses as of 36 months against reported losses as of 24 months following Mack's methodology.

b. (0.5 point)

Identify the chain-ladder method assumption needed for least-squares optimality that can be tested by reviewing the scatter plot created in part a. above and briefly explain whether this assumption has been violated.

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QUESTION 7

TOTAL POINT VALUE: 1.75

LEARNING OBJECTIVE: A2: Estimate parameters and unpaid claims using claims development models related to loss reserving methods such as chain ladder, Cape Cod, chain ladder plus calendar-year effects, and Bornhuetter-Ferguson.

SAMPLE ANSWERS

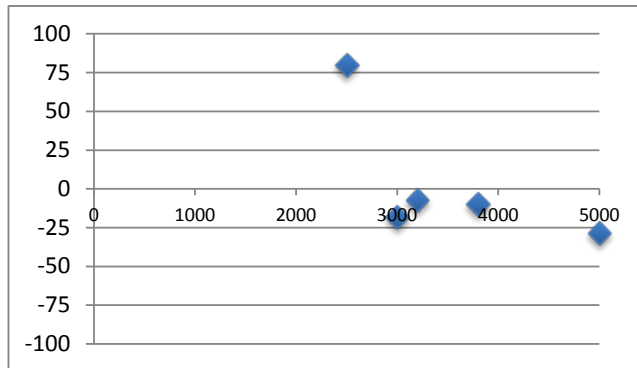
Part a: 1.25 points

Sample Answer 1

$$f = \frac{5000+8000+9000+6000+7000}{3000+5000+2500+3200+3800} = 2$$

$$\frac{\text{Loss @ 36} - \text{Loss @ 24} \times f}{\sqrt{\text{Loss @ 24}}}$$

Rpt Loss @ 24 mo	Residual
3000	-18.257
5000	-28.284
2500	80
3200	-7.071
3800	-9.733



Sample Answer 2

(\$ millions)

$$LDF_{24-36} = \frac{5 + 8 + \dots + 7}{3 + 5 + \dots + 3.8} = 2$$

	(1)	(2)	(3)=2×(1)	(4) = $\frac{(2)-(3)}{\sqrt{(1)}}$
AY	L ₂₄	L ₃₆	LDF×L ₂₄	r _i
9	3	5	6	-0.577
10	...	...	10	-0.894
11	...	...	5	2.5298
12	...	...	6.4	-0.224
13	3.8	7	7.6	-0.308

Chart similar to Sample Answer 1

Part b: 0.5 point

Sample Answer 1

Variance Assumption->variance of the next period's loss is a function of age and cumulative losses to date

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As losses don't appear randomly scattered around 0, (the smallest loss has a large positive residual & all others are negative) this assumption has not been met.

Sample Answer 2

3rd assumption: Variance of incremental loss was a function of loss report-to-date and age
We prefer plot that random scatter around zero
but the plot in (a) was mostly negative and one point highly positive
This violates Mack's 3rd assumption

EXAMINER'S REPORT

Candidates were expected to demonstrate an understanding of how to test Mack's chain-ladder assumptions needed for least squares optimality, in particular the variance assumption. Candidates generally scored well on this question, though some struggled to clearly define Mack's 3rd assumption.

Part a

This part required candidates to produce a scatter plot of weighted residuals. Candidates needed to compute a weighted average LDF and apply the corresponding weighted residual formula to each accident year according to Mack's methodology. The candidates were then expected to sketch a scatter plot of the weighted residuals by reported losses at 24 months.

In order to obtain full credit, candidates needed to properly calculate the weighted average LDF, document the appropriate weighted residual formula, properly calculate the weighed residuals, and provide a labeled scatter plot of the residuals against reported losses at 24 months.

The most common error was using an incorrect weighted residual formula relative to the chosen approach to calculating the LDFs.

Part b

This part required candidates to properly identify the Mack chain-ladder assumption needed for least-squares optimality that can be tested by reviewing the scatter plot from part a. Candidates were also required to explain whether the scatter plot showed that the assumption was violated.

In order to obtain full credit, candidates needed to clearly identify the assumption correctly. Candidates were also required to provide a proper conclusion of the assumption's applicability to the data based on the scatter plot in part a. A reasonable explanation of the candidate's rationale for the conclusion was also required.

Common errors included:

- Not specifying that the variance is of the next observation's cumulative/incremental loss
- Saying that the variance is of the expected loss